

KEATON ZELAZNY

Boston, MA | (440) 214-4312 | zelazny.k@northeastern.edu | linkedin.com/in/keaton-zelazny | keatonzelazny.com

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Engineering and Computer Science

Expected May 2027

GPA: 3.94 / 4.00 | Dean's List | Global Scholars Program: Oakland, CA & London, UK (2023 - 2024)

Relevant Coursework: Embedded Design: Enabling Robotics; Computer Systems; Fundamentals of Digital Design; Fundamentals of Electronics; Computer Architecture and Organization

TECHNICAL SKILLS

Hardware / EDA: SystemVerilog, Quartus Prime, Vivado Design Suite, PSpice, FPGA

Languages: C, C++, Python, Java, MATLAB, (RISC-V, x86) Assembly

Tools: Git, Unix/Linux, GDB, MS Office Suite

TECHNICAL PROJECTS

Single-Cycle RISC-V Processor (*SystemVerilog, Vivado, PYNQ-Z2 FPGA*)

Spring 2026

- Designed and synthesized a RISC-V-compressed-instruction datapath and control unit (program counter, ALU, register file, branch logic) using SystemVerilog under the Vivado Design Suite.
- Wrote custom RISC-V assembly test programs exercising signed multiplication and branch edge cases; verified via SystemVerilog testbench simulation and Vivado VIO hardware debugging.

RL for Bipedal Locomotion (*Python, MuJoCo, Gymnasium, Stable-Baselines3*)

Fall 2025 - Present

- Designed a reference-gait tracking reward function and trained a PPO gait policy for a custom 6-DOF biped.
- Assembling physical prototype (ESP32, I2C servo driver and IMU, six-servo leg assembly) toward sim-to-real deployment.

Analog Front-End Signal Conditioning (*Op-amp filter design, PSpice*)

Spring 2025

- Designed cascaded amplification and filtering stages to recover a clean physiological waveform from raw electrode input.

Six-Legged Walking Robot (*C++, memory-mapped I/O*)

Fall 2024

- Implemented PWM register control and angle/speed-to-duty-cycle conversion for an 18-servo hexapod; added backward walking and directional turning to an alternating tripod gait; cross-compiled for an ARM/Linux FPGA SoC target.

WORK EXPERIENCE

Genezen

Lexington, MA

Automation Engineer Co-op

Jul 2025 - Dec 2025

- Replaced end-of-life sensor transmitters and rebuilt their integration into a live monitoring system, authenticating to a vendor REST API over OAuth2 and scripting real-time data ingestion in Jython.
- Designed parameterized data type definitions and operator display screens, wiring every instance into automatic SQL logging so new sensors required no per-unit configuration.

Integrated Mill Systems

Willoughby, OH

Fabricator

Summer 2023

- Assembled custom control panels for industrial automation systems, building to wiring schematics and panel layout drawings.

INTERESTS

Cooking, NBA basketball, video games, travel, reading, robotics, painting, community engagement.